EXPERIMENT 2

Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
uploaded=files.upload()
df=pd.read_csv('SALES DATA - SALES DATA.csv')
print(df.head())
print(df.isnull().sum())
df['Sales']=(df['Sales'].fillna(df['Sales'].mean))
df.dropna(subset=['Product','Quantity','Region'],inplace=True)
print(df.describe())
product summary=df.groupby('Product').agg({
    'Sales':'sum',
    'Quantity':'sum'
}).reset_index()
print(product_summary)
plt.figure(figsize=(10,6))
plt.bar(product_summary['Product'],product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date']=pd.to_datetime(df['Date'],format='%d-%m-%Y')
sales over time=df.groupby('Date').agg({'Sales':'sum'}).reset index()
plt.figure(figsize=(10,6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()
pivot table=df.pivot table(values='Sales',index='Region',columns='Produ
ct',aggfunc='sum',fill_value=0)
print(pivot_table)
correlation matrix=df.select dtypes(include='number').corr()
print(correlation_matrix)
import seaborn as sns
plt.figure(figsize=(10,6))
sns.heatmap(correlation_matrix,annot=True,cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

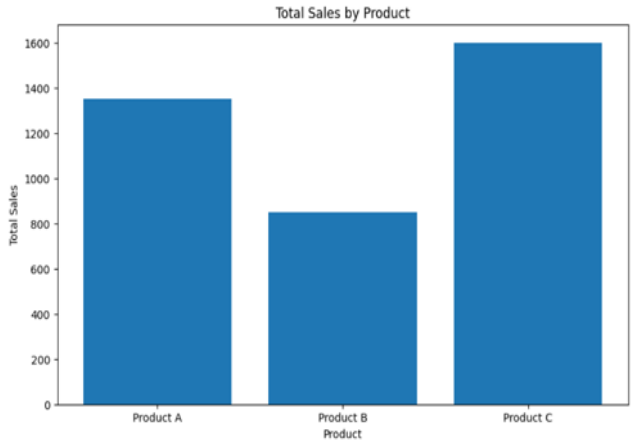
OUTPUT:

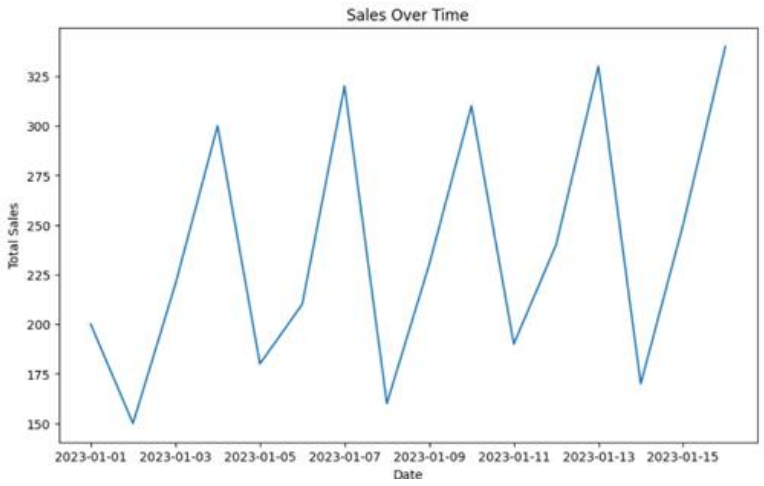
	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A	200	4	North
1	02-01-2023	Product B	150	3	South
2	03-01-2023	Product A	220	5	North
3	04-01-2023	Product C	300	6	East
4	05-01-2023	Product B	180	4	West

```
Date       0
Product    0
Sales      0
Quantity   0
Region     0
dtype: int64
```

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36





Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000

